

Prediction Methodology Standard (PMES)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Operational Layer: Prediction Methodology Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard

Governed Space:

Prediction Methodology

Origin ID: OOF-PGA-PMES

Version: 1.0

Status: Canonical · Open Standard

Effective Date: 6 September 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

1. Canonical Definition

The Prediction Methodology Standard (PMES) governs the establishment, selection, structure, documentation, applicability, traceability, and version governance of the methodology by which a governed predictive frame is intended to be transformed into a prediction.

PMES establishes the methodological basis necessary to distinguish a governed prediction from an unsupported expectation, intuition, assertion, opinion, or prediction-like output.

A Prediction Methodology must be sufficiently defined to establish how predictive inputs, assumptions, logic, models, analytical mechanisms, human judgment, computational processes, aggregation mechanisms, or other applicable predictive methods contribute to prediction formation.

A prediction without a documented methodology is an opinion, not a governed prediction.



Operational Role

PMES is the second Parent Standard within the PGA™ Prediction Lifecycle.

It receives the governed predictive frame established by POPS — Prediction Object & Purpose Standard and governs the methodology intended to produce the prediction.

Its principal architecture transition is:

Governed Prediction Object & Purpose → Governed Prediction Methodology

PMES does not produce the prediction.

It establishes the methodological structure under which prediction formation may subsequently occur.

3. Core Question

By what documented and reproducible methodology is the prediction legitimately constructed?

4. PGA™ Governance Flow

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Current Position: Prediction Methodology

PMES receives the governed predictive frame from POPS and provides governed Prediction Methodology to PBAS — Prediction Basis Standard and subsequent PGA™ governance.

5. Governed Space

PMES governs the Prediction Methodology space.

This includes architecture-level governance of:

predictive method selection; methodological structure; predictive logic; methodological documentation; method applicability; methodological reproducibility; methodological traceability; methodological dependencies; methodological limitations; methodological change; methodology version governance.

PMES governs the methodology used to construct predictions.

It does not determine whether the resulting prediction is correct.

6. Prediction Methodology

A Prediction Methodology is the documented methodological structure through which a governed predictive frame, applicable Prediction Basis, assumptions, and other relevant predictive conditions can be transformed into a prediction.

A methodology may involve:

statistical methods; mathematical models; machine-learning systems; AI models; human forecasting; expert judgment; analytical frameworks; econometric methods; physical models; forecasting algorithms; prediction markets; ensemble mechanisms; simulation-supported processes; hybrid human-machine methods; other documented predictive mechanisms.

The existence of a predictive mechanism does not by itself establish a governed Prediction Methodology.

7. Methodological Identity

A governed Prediction Methodology must possess sufficient identity to remain distinguishable from other methodologies and from materially different versions of itself.

Methodological identity enables subsequent predictive outputs to remain associated with the methodology under which they were produced.

A generic statement such as “AI prediction,” “expert analysis,” “statistical model,” or “market forecast” may describe a methodological class without sufficiently identifying the methodology actually applied.

Method Category ≠ Governed Prediction Methodology

8. Method Selection

The selected predictive method must remain identifiable and relevant to the governed predictive frame.

PMES requires that method selection be methodologically justifiable rather than assumed merely because a method is available, popular, computationally powerful, historically successful, or technically capable of producing an output.

Method Availability ≠ Method Applicability

Predictive Capability ≠ Methodological Suitability

Detailed method-selection mechanics belong to PMSM — Predictive Method Selection Module .

9. Methodological Structure

A governed Prediction Methodology must possess sufficient structure to explain how predictive treatment proceeds from governed predictive conditions toward prediction formation.

The structure must make material methodological components and their relationships identifiable.

Methodological complexity is not itself evidence of methodological quality.

Likewise, methodological simplicity does not automatically imply weakness.

Methodological Complexity ≠ Methodological Validity

PMES governs whether the methodology is sufficiently structured, not whether it is unnecessarily complex.

10. Predictive Logic

The material logic by which the methodology transforms predictive inputs and conditions toward a predictive result must be sufficiently identifiable.

A methodology must not rely entirely upon unexplained transformation where that transformation prevents the predictive process from being methodologically understood, governed, traced, or reproduced to the degree required by its context.

Predictive logic may be computational, statistical, analytical, judgment-based, market-based, hybrid, or otherwise methodologically defined.

Prediction Output ≠ Documented Predictive Logic

11. Methodological Documentation

A governed Prediction Methodology must be sufficiently documented to preserve its identity, structure, logic, applicability, material dependencies, and version.

Documentation must allow the methodology associated with a prediction to remain distinguishable from undocumented reasoning or retrospective explanation.

Documentation may vary in form according to the methodology and predictive context.

The governance requirement is methodological sufficiency, not documentation volume.

Documentation Volume ≠ Methodological Governance Quality

12. Methodological Reproducibility

A governed methodology must provide sufficient methodological continuity for the process by which a prediction was produced to be reconstructable or reproducible to the degree reasonably applicable to the method.

Reproducibility does not require every prediction to produce an identical result under every repetition.

Human judgment, stochastic systems, dynamic models, markets, and probabilistic processes may legitimately produce variation.

The requirement is that the methodology itself remain sufficiently identifiable.

Methodological Reproducibility $\neq$ Identical Predictive Output

13. Method Applicability

A methodology must be applicable to the Prediction Object, Predictive Purpose, Prediction Target, Prediction Horizon, and Predictive Boundary it is intended to serve.

A method that is legitimate for one predictive context may be inappropriate for another.

Historical success, technical sophistication, model scale, institutional reputation, or market adoption does not independently establish applicability.

Methodological Reputation $\neq$ Method Applicability

Past Performance $\neq$ Universal Method Applicability

14. Methodological Fit

PMES requires sufficient alignment between the methodology and the predictive frame established by POPS.

Methodological fit may be affected by:

Prediction Object; Predictive Purpose; Prediction Target; Prediction Horizon; Predictive Boundary; nature of available predictive information; required predictive resolution; applicable conditions; methodological limitations.

PMES governs whether this relationship is methodologically coherent.

It does not govern the sufficiency of the Prediction Basis itself.

15. Methodological Dependencies

A Prediction Methodology may depend upon models, datasets, external systems, human judgment, simulations, market mechanisms, software, algorithms, instruments, analytical assumptions, or other components.

Material methodological dependencies must remain identifiable where their presence affects the meaning, reproducibility, applicability, or continuity of the methodology.

Identification of a dependency does not transfer governance ownership of that dependency to PMES.

16. Human and Machine Methodologies

PGA™ does not privilege human or machine prediction by default.

Human forecasting, AI prediction, algorithmic forecasting, expert judgment, collective mechanisms, and hybrid approaches may all constitute legitimate methodological components when sufficiently governed.

The relevant question is not whether the predictor is human or machine.

The relevant question is whether the predictive methodology is sufficiently defined.

Human Judgment ≠ Ungoverned by Default

Machine Output ≠ Governed by Default

17. Hybrid Prediction Methodologies

A methodology may combine multiple predictive mechanisms.

Examples may include human judgment with statistical models, AI outputs with expert review, simulations with analytical forecasting, or multiple models operating within an ensemble.

Where methods are combined, their material methodological relationship must remain identifiable.

Combining governed methods does not automatically produce a governed combined methodology.

Multiple Methods ≠ Governed Hybrid Methodology

18. Model and Method Separation

A predictive model may form part of a Prediction Methodology without constituting the entire methodology.

The methodology may additionally govern how the model is selected, configured, supplied with predictive inputs, interpreted, combined with other mechanisms, or used within the predictive process.

Therefore:

Prediction Model ≠ Prediction Methodology

and:

Model Output ≠ Governed Prediction

This distinction is especially important in AI and automated predictive environments.

19. Simulation and Prediction Separation

Simulation may contribute information, scenarios, distributions, pathways, or other outputs to a Prediction Methodology.

Simulation does not automatically constitute prediction.

Where simulation is used, SIMULOS® retains governance ownership of simulation, while PGA™ governs how simulation- derived outputs participate in prediction methodology where applicable.

Simulation Output ≠ Prediction

Simulation Methodology ≠ Prediction Methodology

Shared methodological components do not erase architecture boundaries.

20. Collective and Market-Based Methodologies

Prediction markets, forecasting tournaments, expert panels, ensembles, multi-agent systems, and other collective mechanisms may participate in Prediction Methodology.

Their existence does not automatically establish that the resulting collective signal is a governed prediction.

Where multiple predictive sources are combined, subsequent collective prediction governance remains subject to CPGS — Collective Prediction Governance Standard .

Market Mechanism ≠ Governed Prediction Methodology by Default

Collective Signal ≠ Governed Prediction

21. Methodology and Prediction Basis Separation

Prediction Methodology governs how predictive treatment is performed .

Prediction Basis governs what information, data, observations, evidence references, and signals form the basis of the prediction .

These spaces must remain distinct.

Prediction Methodology $\neq$ Prediction Basis

A well-documented methodology cannot compensate for an insufficient Prediction Basis, and a strong Prediction Basis does not establish a governed methodology.

22. Methodology and Assumption Separation

Prediction Methodology and Prediction Assumptions must remain distinguishable.

The methodology defines how prediction is constructed.

Assumptions define conditions treated as holding for that predictive process.

Assumption governance belongs to PAS — Prediction Assumption Standard .

Prediction Methodology $\neq$ Prediction Assumption

A methodology must not conceal material assumptions by representing them as methodological facts.

23. Methodology and Prediction Formation Separation

PMES governs the methodology under which a prediction can be constructed.

It does not construct the specific prediction.

That lifecycle function belongs to PFOS — Prediction Formation Standard .

Prediction Methodology $\neq$ Prediction Formation

Defined Methodology $\neq$ Formed Prediction

This separation allows one methodology to support multiple predictive instances while preserving the identity of each prediction.

24. Methodology and Probability Separation

A methodology may produce or contribute to a probability estimate.

PMES does not determine how probability, confidence, uncertainty, ranges, or conditionality are ultimately represented.

That governance belongs to PPUS — Prediction Probability & Uncertainty Standard .

Methodological Output $\neq$ Governed Probability Representation

The origin of probability-like values must remain methodologically traceable.

25. Probability Provenance

Where a methodology produces, transforms, aggregates, or incorporates a probability-like representation, its methodological origin must remain identifiable.

A probability may originate from:

statistical estimation; model output; expert judgment; human forecasting; AI forecasting; ensemble methodology; collective aggregation; prediction-market pricing; simulation-supported estimation; hybrid methodology; another documented mechanism.

Identical numerical values may carry materially different predictive meaning depending upon their methodological origin.

Same Probability Value $\neq$ Same Predictive Meaning

PMES establishes methodological provenance; PPUS governs probability and uncertainty representation.

26. Methodological Transparency

A methodology must provide sufficient transparency for its governed role to be understood.

Transparency does not require unrestricted disclosure of proprietary code, protected intellectual property, confidential information, or every internal computational operation.

It requires sufficient methodological visibility to establish what method was used, how it functions at the governance- relevant level, and what material limitations or dependencies affect its predictive role.

Methodological Transparency $\neq$ Unlimited Technical Disclosure

27. Methodological Limitations

Material limitations of the Prediction Methodology must remain identifiable.

A methodology may be valid within a defined predictive context while unsuitable outside it.

Limitations may affect applicability, interpretability, resolution, temporal reach, domain transfer, dependency, or other methodological characteristics.

A limitation does not automatically invalidate the methodology.

Methodological Limitation $\neq$ Methodological Failure

Failure to recognize a material limitation may, however, create a governance gap.

28. Methodological Change

Prediction Methodologies may evolve.

A material methodological change must remain distinguishable from continuation of the original methodology.

Changes may concern:

method selection; methodological structure; predictive logic; model configuration; aggregation mechanism; material dependencies; analytical procedure; other methodology-defining components.

Changed Methodology $\neq$ Same Methodology by Default

Material change may require a new governed methodology version.

29. Methodology Version Governance

A governed Prediction Methodology must preserve sufficient version identity to determine which methodology governed a particular prediction.

Later methodological improvement must not retrospectively alter the methodology under which an earlier prediction was produced.

Current Methodology $\neq$ Historical Methodology

Methodology Update $\neq$ Historical Replacement

This enables legitimate later reconstruction, comparison, and performance interpretation.

30. Methodological Continuity

Methodological continuity requires the relationship between methodology versions and predictive instances to remain identifiable over time.

Continuity does not prohibit improvement.

It ensures that evolution does not erase the methodological conditions under which historical predictions were formed.

Detailed methodology version mechanics belong to MVGM — Methodology Version Governance Module .

31. Methodological Sufficiency

A Prediction Methodology is sufficiently governed when it is:

identifiable; appropriately selected; structurally defined; logically documented; applicable to the governed predictive frame; sufficiently reproducible or reconstructable; traceable to material dependencies; transparent at the governance-relevant level; bounded by identifiable limitations; versioned where material change occurs.

Methodological sufficiency does not establish predictive correctness.

Governed Methodology $\neq$ Correct Prediction

32. Governance Requirements

PMES requires that:

1. a Prediction Methodology be explicitly identifiable; 2. method selection be methodologically relevant to the governed predictive frame; 3. material methodological structure be sufficiently defined; 4. predictive logic be sufficiently documented; 5. methodology applicability be established; 6. material methodological dependencies remain identifiable; 7. methodology remain distinguishable from Prediction Basis and assumptions; 8. model output not be treated as a governed prediction solely because a model produced it; 9. probability-like outputs retain methodological provenance; 10. material methodological limitations remain identifiable; 11. material methodological changes remain traceable; 12. methodology versions remain distinguishable; 13. historical predictions remain associated with the methodology version under which they were formed.

33. Governance Logic

PMES applies the architecture-level governance sequence:

Governed Prediction Object & Purpose

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Predictive Method Selection

↓

Methodological Structure Definition

↓

Predictive Logic Documentation

↓

Method Applicability

↓

Methodology Version Governance

↓

Governed Prediction Methodology

↓

PBAS — Prediction Basis Standard

Where methodological sufficiency is not established:

Methodological Governance Gap → Clarification / Reassessment → Methodology Governance

A prediction must not acquire governed status merely because an output can be generated.

34. Governance Outputs

PMES may produce:

Governed Prediction Methodology;

methodology identity; selected predictive method; methodological structure reference; predictive logic reference; applicability state; methodological dependency context; methodological limitation context; methodology version; methodological continuity reference; unresolved methodological governance requirements.

The principal governed output is:

Governed Prediction Methodology

This becomes a foundational input into subsequent PGA™ governance.

35. Internal Module Architecture

PMES is operationalized through five Core Modules:

PMSM — Predictive Method Selection Module

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MSDM — Methodological Structure Definition Module

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PLDM — Predictive Logic Documentation Module

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MAPM — Method Applicability Module

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MVGM — Methodology Version Governance Module

Each module governs a distinct operational mechanism within the Prediction Methodology space.

36. Critical Distinctions

Prediction Methodology ≠ Prediction

Prediction Methodology ≠ Prediction Basis

Prediction Methodology ≠ Prediction Assumption

Prediction Methodology ≠ Prediction Formation

Prediction Model ≠ Prediction Methodology

Model Output ≠ Governed Prediction

Simulation Output ≠ Prediction

Method Category ≠ Governed Prediction Methodology

Method Availability ≠ Method Applicability

Predictive Capability ≠ Methodological Suitability
Methodological Complexity ≠ Methodological Validity
Methodological Reproducibility ≠ Identical Predictive Output
Multiple Methods ≠ Governed Hybrid Methodology
Same Probability Value ≠ Same Predictive Meaning
Methodological Transparency ≠ Unlimited Technical Disclosure
Methodological Limitation ≠ Methodological Failure
Changed Methodology ≠ Same Methodology by Default
Methodology Update ≠ Historical Replacement
Governed Methodology ≠ Correct Prediction

37. Cross-Architecture Boundaries

PMES governs Prediction Methodology within PGA™.

It does not assume governance ownership of:

simulation governance under SIMULOS® ; Operational Reality under ORA™ ; evidence governance under ADIT® ; validation governance under VALIDOS™ ; cognition governance under CLIA® ; AI behavioral governance under AIG® ; autonomous-system governance under ASGA™ ; integrity governance under INTEGROS® ; accountability governance under AGA™ .

Outputs from these architectures may participate in predictive methodology where applicable without transferring their underlying governance ownership to PGA™.

38. Standard Boundary

PMES governs the selection, identity, structure, documentation, logic, applicability, methodological provenance, limitations, change, versioning, and continuity of the methodology used to construct a prediction. It does not govern the sufficiency of the Prediction Basis, predictive assumptions, formation of the specific prediction, probability and uncertainty representation, feasibility, collective prediction governance, communication and reliance, revision, Operational Reality determination, predictive performance, or Prediction Record continuity.

PMES governs how prediction is methodologically constructed .

It does not determine what the future will be .

Canonical Closing Statement

A predictive output becomes governable not because it appears precise, sophisticated, or plausible, but because the methodology by which it was produced can be identified, understood, bounded, and traced.

A prediction without a documented methodology is an opinion, not a governed prediction.

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Prediction Methodology Standard

Architecture:

PGA™ — Prediction Governance Architecture

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Canonical Definition: The Prediction Methodology Standard governs the selection, structure, documentation, applicability, traceability, and version governance of the methodology by which a governed predictive frame is transformed into a prediction.

Governed Space:

Prediction Methodology

Governance Flow:

Prediction Object & Purpose → → Prediction Methodology → Prediction Basis → Prediction Assumptions
→ Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where
applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance →
Prediction Record & Continuity

A prediction without a documented methodology is an opinion, not a governed prediction.

→ View Standard

About standard

Governed Links

[About Prediction Methodology Standard \(PMES\).](#)

[Prediction Methodology Specification Module \(PMSM\).](#)

[Methodological Structure Definition Module \(MSDM\).](#)

[Prediction Logic Documentation Module \(PLDM\).](#)

[Methodology Application Module \(MAPM\).](#)

[Methodology Version Governance Module \(MVGM\).](#)

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